PHAM MINH HIEU

FRESHER PYTHON BACKEND DEVELOPER

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PROFILE

Recent Information Technology graduate (HUTECH, 2025), now pursuing a Master's degree in IT. Built and tested Python applications in a Master's thesis and academic projects using FastAPI, OpenCV and PyTorch. Practised PostgreSQL, SQLAlchemy and Docker through coursework and self-study. Seeking a Fresher Python Backend Developer role to gain production experience in a professional team.

TECHNICAL SKILLS

Programming Languages:	Python, SQL, JavaScript, HTML/CSS
Backend & API:	FastAPI, Uvicorn, RESTful API, HTTP/JSON, Multithreading
AI & Computer Vision:	OpenCV, PyTorch, ONNX Runtime
Data Processing:	NumPy, pandas, SciPy, scikit-learn
Testing & Tools:	unittest, Git, GitHub, Conda, VS Code, Postman
Familiar With:	PostgreSQL, SQLAlchemy, Pydantic, JWT/OAuth2, Pytest, Docker, Docker Compose, Linux, Streamlit

PROJECTS

Video-Based Emotion Recognition for Public-Service Transactions — Master's thesis **2026 – Present**

Python, FastAPI, OpenCV, ONNX Runtime, PyTorch, pandas, SciPy, HTML/CSS/JavaScript

- Built a FastAPI backend (35+ REST endpoints) with a background job queue for video upload, processing, evaluation and analysis, serving a single-page HTML/CSS/JS dashboard.
- Implemented an OpenCV video pipeline: face detection (YuNet), alignment, geometry-only tracking (Kalman filter + IoU), and per-frame 7-class emotion inference with pretrained models.
- Benchmarked 6 open-source emotion-recognition libraries on FER2013 PrivateTest (best: 72.3% accuracy) and under 7 image-corruption types × 5 severity levels.
- Aggregated frame-level predictions into 130+ session-level features and tested their link to citizen satisfaction scores (Spearman, Benjamini–Hochberg FDR).
- Sped up face detection 2.44× with no loss in face coverage through adaptive frame downscaling and selective frame decoding.
- Wrote 100+ automated regression tests (unittest) against a live Uvicorn server on synthetic data; kept all processing local, with no identity recognition, to protect personal data.

Food Search and Selection Website — Academic team project **Mar 2023 – Jun 2023**

MVC architecture, SQL database, Git/GitHub

- Developed data-processing functions and database connectivity following the MVC architecture.
- Built a responsive user interface for searching and selecting dishes.
- Worked with teammates on feature development and managed source code with Git and GitHub.

EDUCATION

Master of Information Technology — HUTECH University (Ho Chi Minh City University of Technology) **Expected 2027**

Thesis: Research and application of deep learning for emotion recognition in public administrative transaction videos

Bachelor of Information Technology — HUTECH University (Ho Chi Minh City University of Technology) **2025**

LANGUAGES

- English: B1 certificate

ACHIEVEMENTS & ACTIVITIES

- Top 3, CODE YOUR FUTURE competition organized by CJ